

Human Genetics IV: Multifactorial and complex disorders

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Acknowledgement

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Learning Objectives

- Describe the principles of inherited multifactorial disorders
- Relate the threshold model to disease expression
- Recognize the contribution of environmental and genetic factors to complex disorders
- Differentiate between qualitative and quantitative traits
- Identify the factors needed to determine empirical risk
- Predict the role of multifactorial traits in common complex diseases such as diabetes, hypertension, and cancer
- Compare and contrast tumor suppressor genes to proto-onco/oncogenes and their roles in cancer

References

- Thompson and Thompson Genetics in Medicine
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/browse/book/3-s2.0-C2009059798X>
- Medical Genetics
<https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780323597371000060>
- Medical Genetics: An integrated approach
<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/book.aspx?bookid=2247>

Complex-USA level 1 blueprint-reference only

Information from this presentation relates to:

- REQUIRED ELEMENT 3.1
- FOUNDATIONAL BIOMEDICAL SCIENCES KNOWLEDGE BASE
 - the molecular, biochemical, tissue, and cellular bases of health and disease.
 - medical genetics.
- COMMUNITY HEALTH AND PATIENT PRESENTATIONS RELATED TO WELLNESS
 - genetic screening
- PATIENT PRESENTATIONS RELATED TO HUMAN DEVELOPMENT, REPRODUCTION, AND SEXUALITY
 - congenital/genetic anomalies (eg, Down syndrome, cystic fibrosis, craniosynostosis)
 - dysmorphic features and genetic syndrome stigmata

From the USMLE Step 1 blueprint-reference only

Information from this presentation relates to:

- General Principles of Foundational Science Biochemistry and molecular biology
 - Gene expression: DNA structure, replication, exchange, and epigenetics (eg, imprinting, X-activation, DNA methylation)
- Human development and genetics
 - Principles of pedigree analysis inheritance patterns occurrence and recurrence risk determination
 - Population genetics: Hardy-Weinberg law, founder effects, mutation-selection equilibrium
 - Genetic testing and counseling Genetic mechanisms (eg, penetrance, genetic heterogeneity)

From the USMLE Step 1 blueprint-reference only

Information from this presentation relates to:

- Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis
 - Genetic metabolic and developmental disorders multifactorial: VATER syndrome, association syndromes large genomic changes: Beckwith-Wiedemann syndrome; Down syndrome; Prader- Willi syndrome
 - enzymatic/metabolic: alpha-1 antitrypsin deficiency; porphyria; inborn errors of metabolism (eg, maple syrup urine disease, diseases involving urea cycle); storage diseases (eg, Fabry disease, Tay-Sachs disease, glycogen storage disease, mucopolysaccharidoses)
 - structural protein disorders: amyloidosis; Ehlers-Danlos syndrome; immotile cilia syndrome (Kartagener syndrome; primary ciliary dyskinesia); Marfan syndrome
 - intracellular/extracellular transport receptors: cystic fibrosis; hemochromatosis; Wilson disease
 - triplet repeat/RNA disorders: fragile X syndrome

Comparison of genetic disease prevalence

Type	Incidence at Birth	Prevalence at Age 25	Population Prevalence
	(per 1000)	(per 1000)	(per 1000)
Disorders due to genome and chromosome mutations	6	1.8	3.8
Disorders due to single-gene mutations	10	3.6	20
Disorders with multifactorial inheritance	≈50	≈50	≈600



Multifactorial inheritance

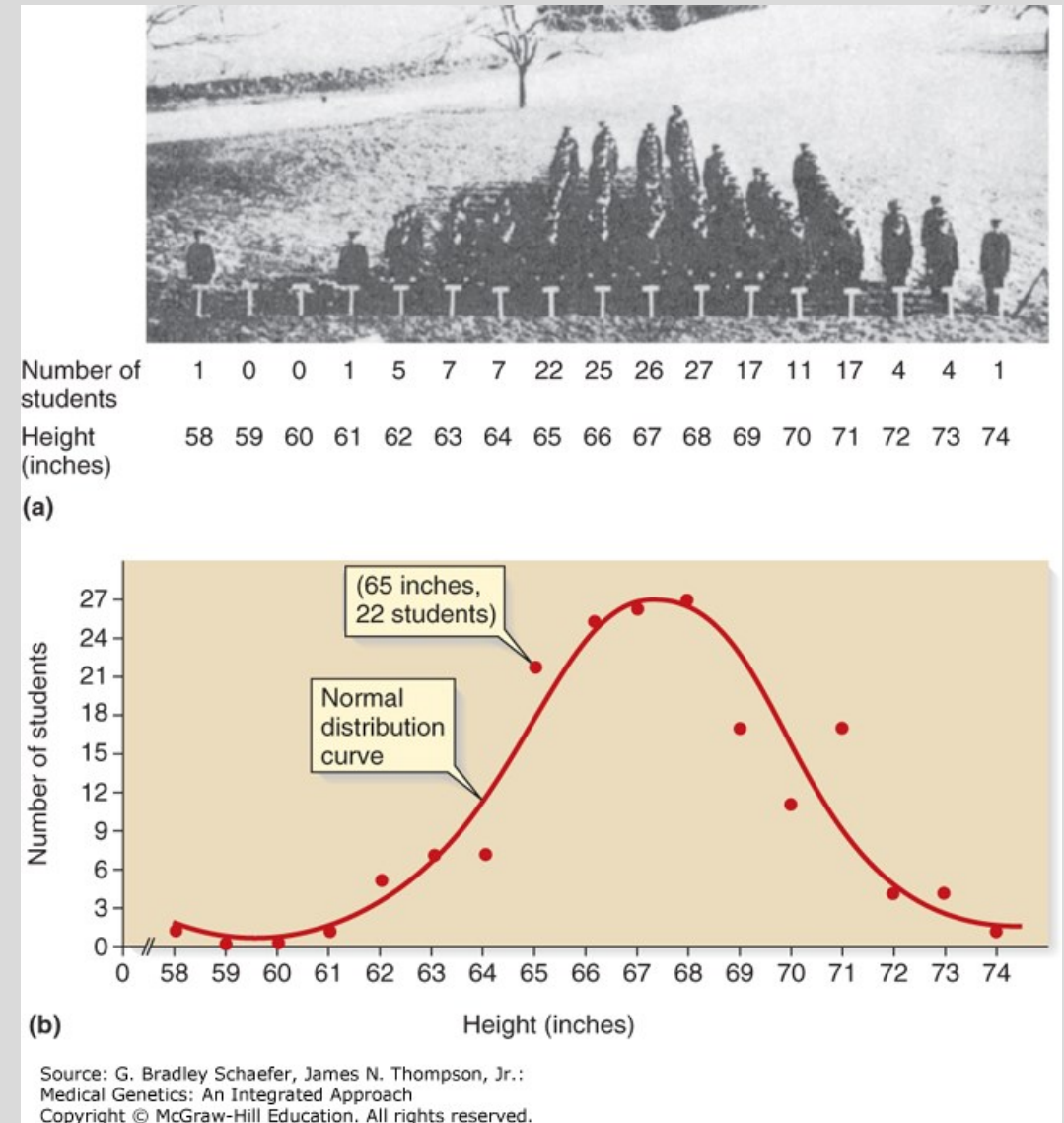
- **Complex interaction between genes and environment**
 - Multifactorial inheritance involves the interplay between multiple genes and environmental factors.
 - Examples include conditions like heart disease, diabetes, and certain cancers.
- **Genetic variability with no uni-factorial mechanism**
 - Unlike single-gene disorders, multifactorial traits do not follow a simple inheritance pattern.
 - The variability in genetic contribution makes it challenging to predict the exact risk.
- **Family studies indicate increased risk for relatives**
 - Family studies show that relatives of affected individuals have a higher risk of developing the same condition.
 - The closer the genetic relationship, the higher the risk.
- **Environmental factors have significant influence**
 - Lifestyle, diet, and exposure to toxins can significantly impact the expression of multifactorial traits.
 - Preventive measures and lifestyle changes can reduce the risk of developing these conditions.

Principles of multifactorial inheritance

1. Segregation does not fit Mendelian inheritance
2. Recurrence risk increases with family members compared to general pop.
3. Inverse relationship between recurrence risk and distance of family members
4. Increased risk with increased severity of condition
5. Many multifactorial conditions display sex bias
6. If condition shows sex bias, relatives of less affected sex at higher risk

Quantitative trait example: Height

- Height is a classic example of quantitative trait.
 - Shows continuous variation in a population
- Polygenic in nature – influenced by many genetic loci, each contributing in portions
 - Multiple genes involved
 - Complex inheritance
- Environmental influence – Nutrition, childhood health



Qualitative vs. Quantitative traits

Qualitative

Quantitative

ABO blood type

Blood pressure

BMI



Diabetes

Height

Can you guess which trait goes under which column?

Threshold model in genetic variations

- Liability distribution
- Genetic and Environmental factors
- Discrete outcomes
- Threshold of liability
- Can have sex bias

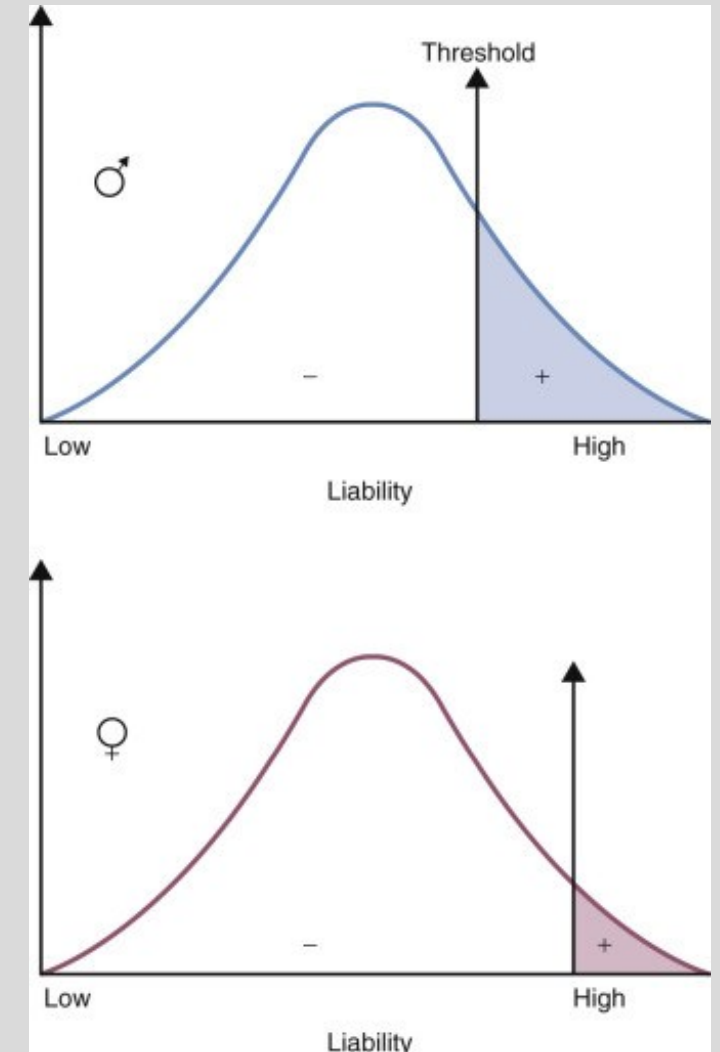
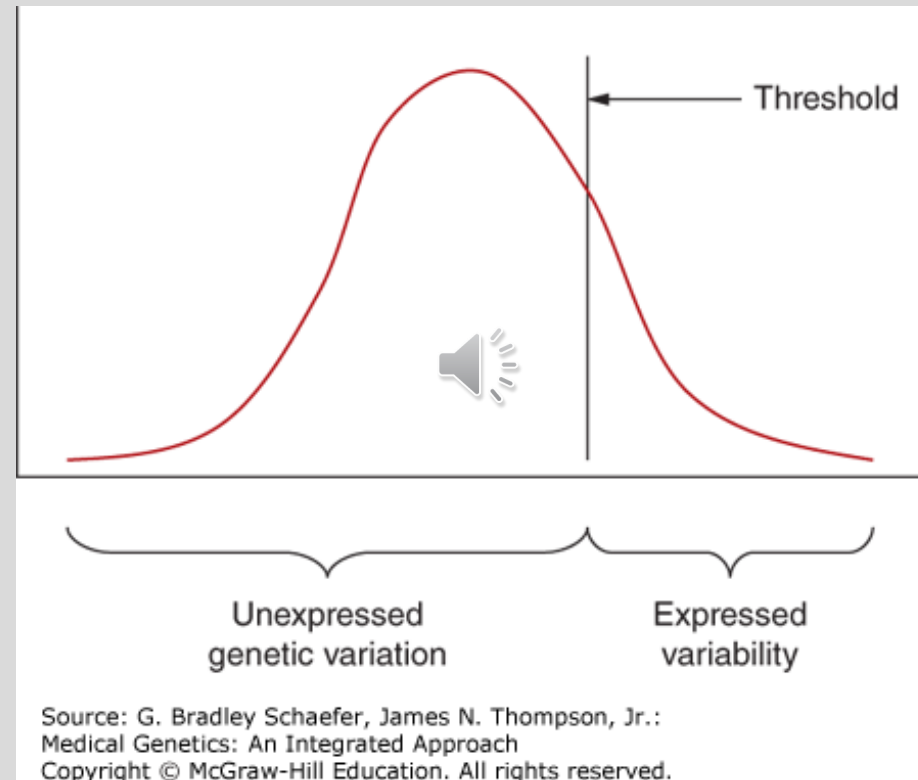
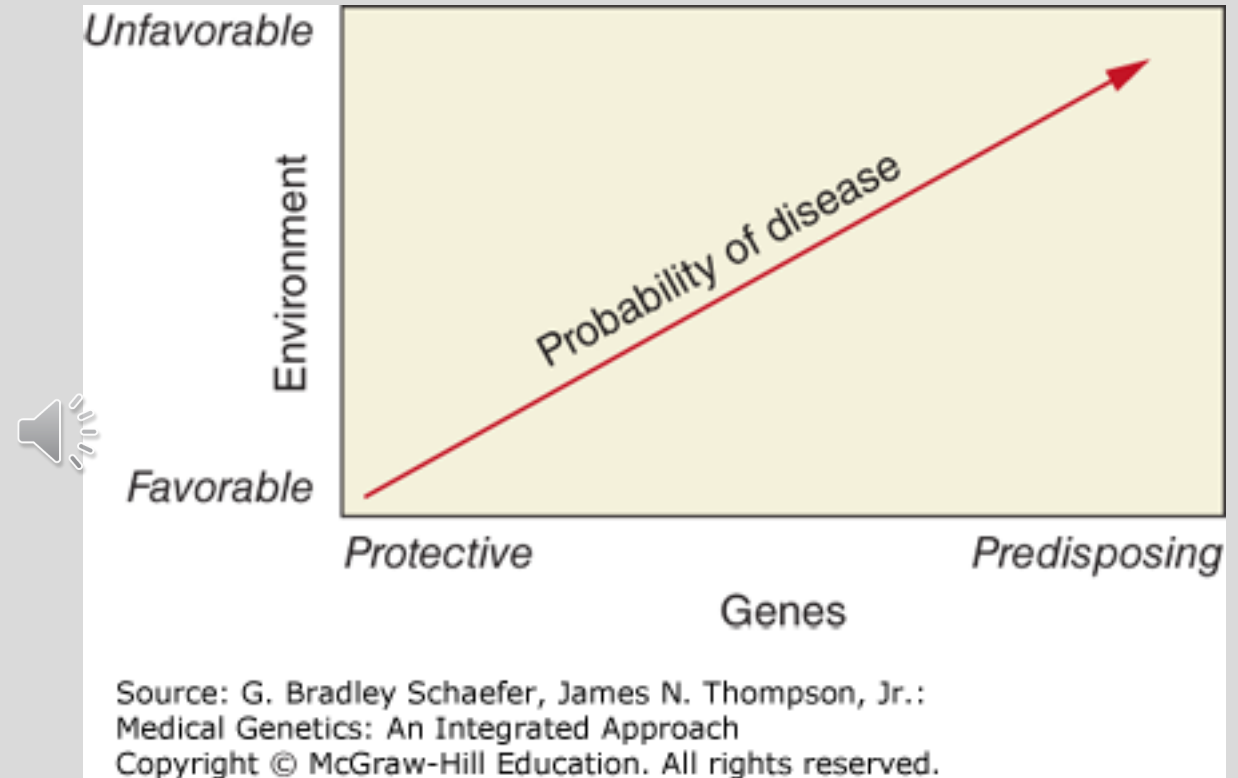


Fig. 12-2 Emery's Elements of Medical Geneti

Example: neural tube defects

- Spina bifida and anencephaly
- Genetic and environmental factors
 - Folic acid intake
- Moms with FA deficiency ≠ NTD
- Moms taking FA can still result in NTD



Condition	After 1 child	After 2 Children	After 3 Children
Spina Bifida	2-5%	5-10%	Upto 10%
Anencephaly	2-5%	5-10%	Upto 10%

Empirical risk

- Calculating exact recurrence risk not possible
- Risk based on observable data
- Disease specific
- Based on collection of family studies
- Also specific to population



Examples: congenital malformations

- Cleft lip/palate

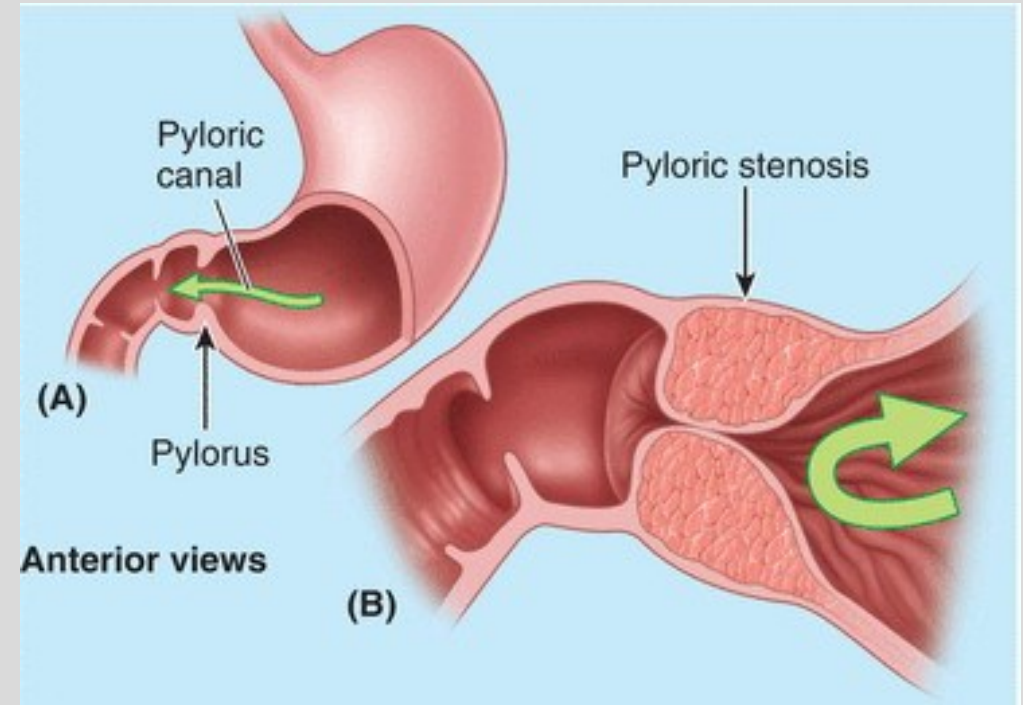
	Risk for CL(P) (%)		
Affected Relatives	No. of Affected Parents		
	0	1	2
None	0.1	3	34
One sibling	3	11	40
Two siblings	8	19	45
One sibling and one second-degree relative	6	16	43
One sibling and one third-degree relative	4	14	44



Table 8-3, Thompson & Thompson Genetics in Medicine, 8th Ed.
Fig. 9-4 Emery's Elements of Medical Genetics

Pyloric stenosis

- Gender bias
- Male with malformation
 - Son: 5.5% risk
 - Daughter: 2.4% risk
- Female with malformation
 - Son: 19.4% risk
 - Daughter: 7.3% risk



RELATIVES	MALE PROBANDS		FEMALE PROBANDS	
	LONDON	BELFAST	LONDON	BELFAST
Brothers	3.8	9.6	9.2	12.5
Sisters	2.7	3.0	3.8	3.8

Fig. B5.9 Moore Clinically Oriented Anatomy 8th edition
Emery's Elements of Medical Genetics

Hypertension

- Systolic >130 mmHg; diastolic >80 mmHg
- Gaussian distribution
- Heritability for BP between 30-50%
- Small % of hypertension due to single-gene mutations (Liddle syndrome) 
- Heritability penetrance $<100\%$ so other factors involved
- **Genetic factors**
 - Polygenic in nature
 - Numerous loci
 - Epigenetic factors
 - Rare single gene mutations
- **Environmental & Lifestyle**
 - Diet and Exercise
 - Stress & Socioeconomic

Diabetes-Type II

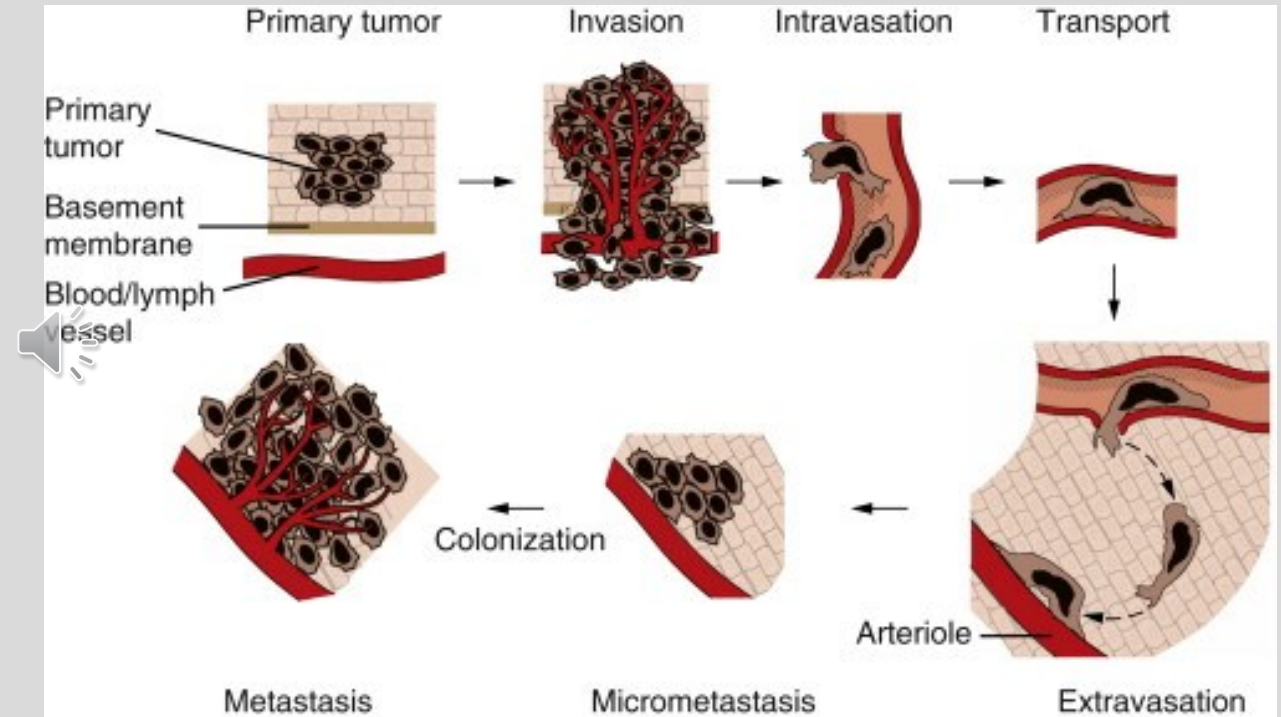
- Insulin resistance
- 90% of all diabetic cases
- 2 major risk factors:
 1. Family history
 2. Obesity-increases insulin resistance
- >70 loci contribute to susceptibility
- Variants of *TCF7L2* identified in all populations studied



FEATURE	TYPE 1 DIABETES	TYPE 2 DIABETES
Age of onset	Usually <40 yr	Usually >40 yr
Insulin production	None	Partial
Insulin resistance	No	Yes
Autoimmunity	Yes	No
Obesity	Not common	Common
MZ twin concordance	0.35-0.50	0.9
Sibling recurrence risk	1-6%	15-40%

Cancer-genetics

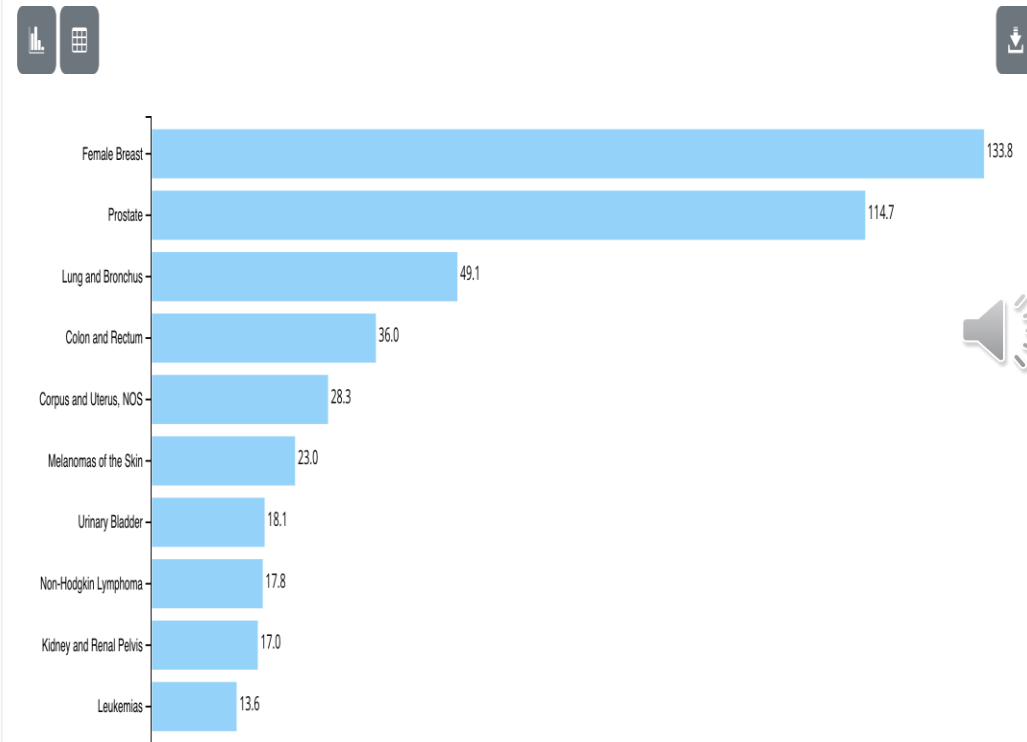
- Genetic factors
 - Polygenic and complex interplay of genes
 - Numerous loci in genome
 - Sometimes autosomal dominant – BRCA 1 & 2
 - Syndrome based – Lynch
- Environmental & Lifestyle
 - Carcinogens
 - Poor diet & lifestyle
 - Stress and Socioeconomic



Cancer-rates - CDC data

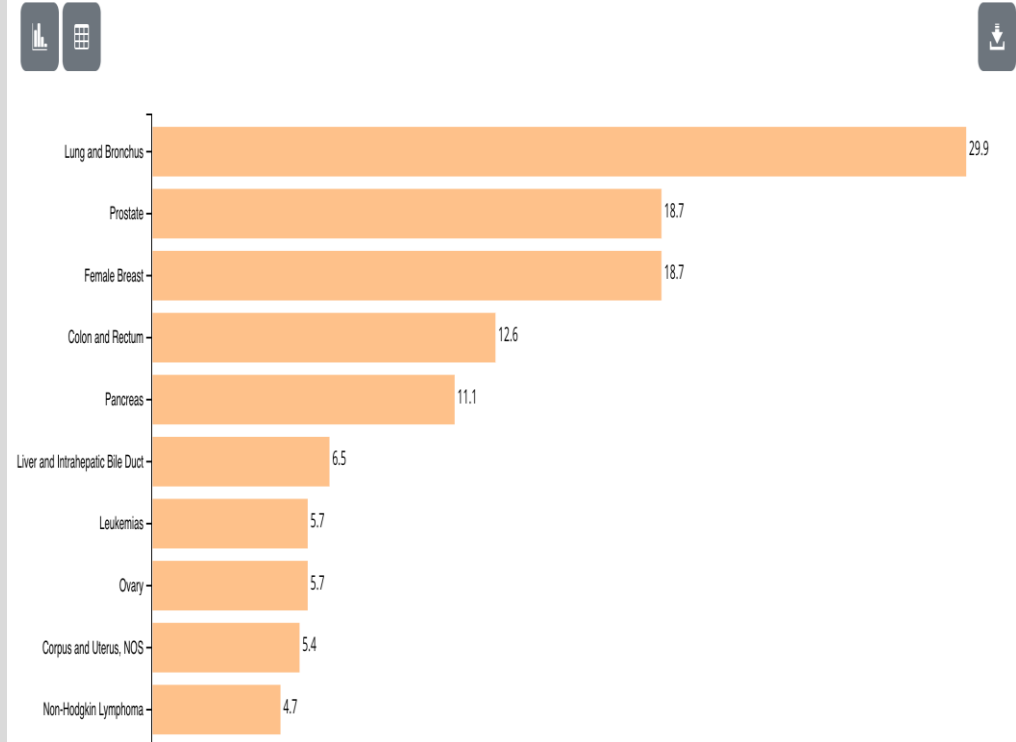
Top 10 Cancers by Rates of New Cancer Cases

United States, 2021, All Races and Ethnicities, Male and Female
Rate per 100,000 people



Top 10 Cancers by Rates of Cancer Deaths

United States, 2022, All Races and Ethnicities, Male and Female
Rate per 100,000 people



Note: In **2021**, the latest year for which incidence data are available, in the **United States**, **1,777,566 new cases of cancer** were reported. For every 100,000 **people**, **439 new cancer** cases were reported.

In **2022**, the latest year for which mortality data are available, in the **United States**, **608,366 people died** of cancer. For every 100,000 **people**, **142 people died** of cancer.

Cancer is the second leading cause of death in the United States, exceeded only by heart disease.

NCI SEER – <https://seer.cancer.gov/>

<https://gis.cdc.gov/Cancer/USCS/DataViz.html>

Oncogenes

- Proto-oncogene: normal gene involved in cell cycle/proliferation
 - Growth factors, transcription factors, receptors, signal transduction
- Oncogene: mutated proto-oncogene
 - Increased activity leading to increased proliferation
 - What can lead to increased activity?
 - Gain-of-function
 - Gene dosage

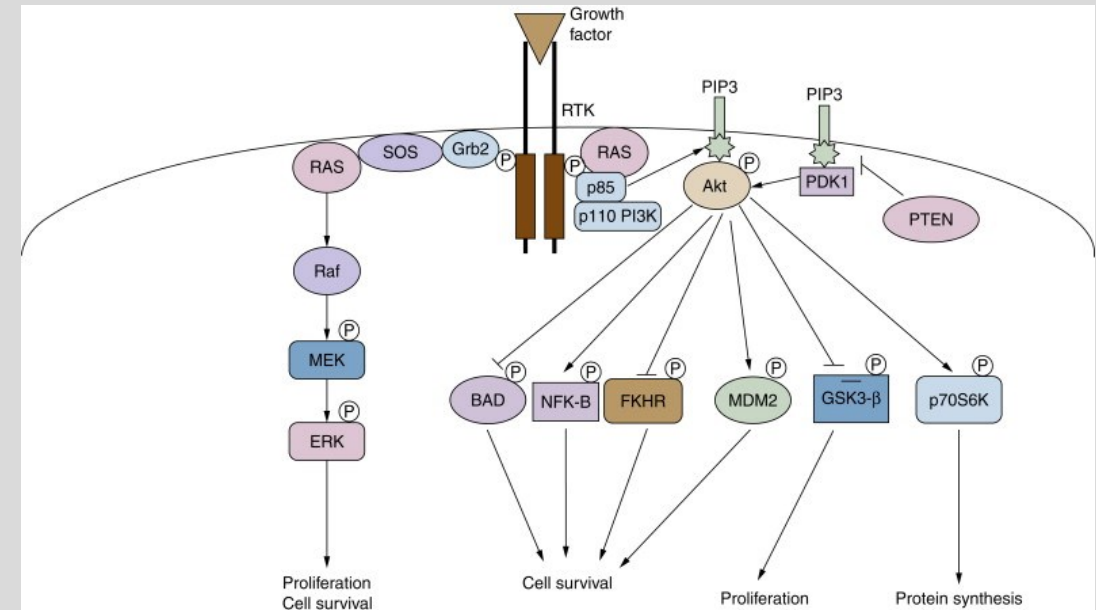


Fig 2-1, The Molecular Basis of Cancer, 4th Ed

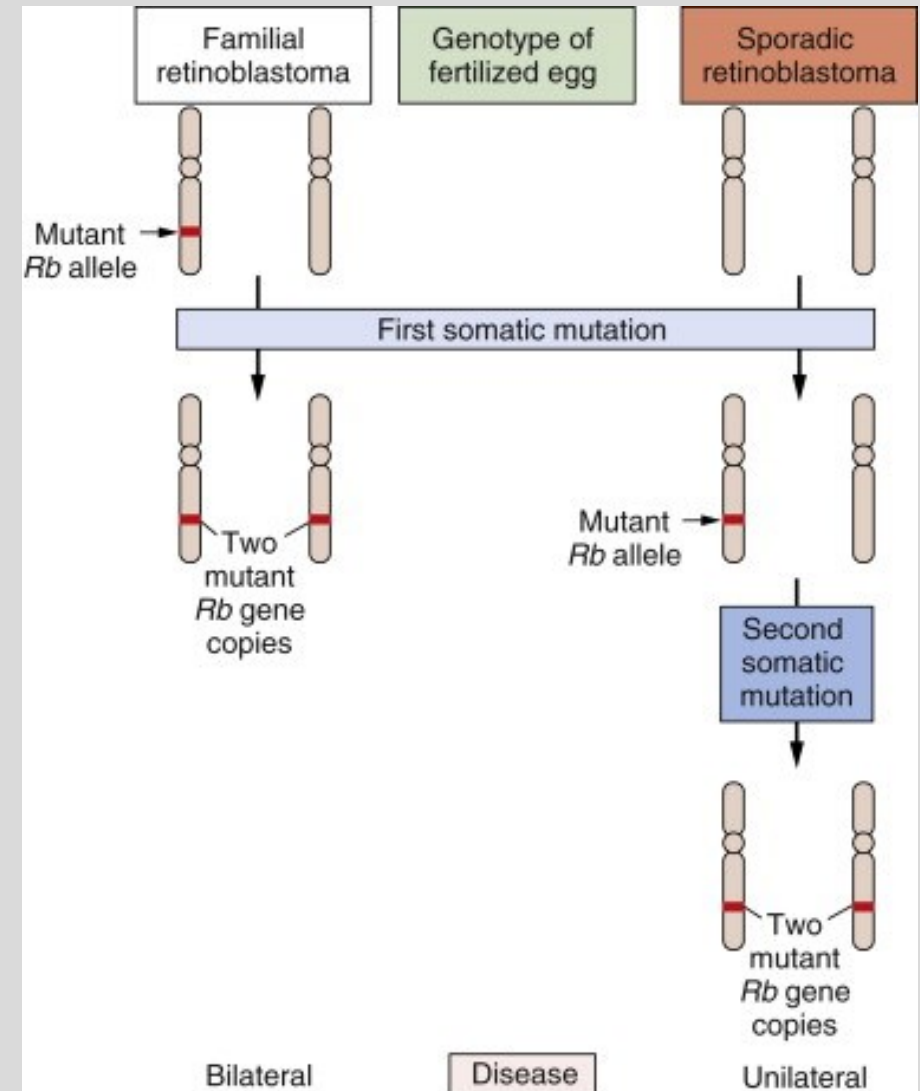
Tumor suppressor genes

- Tumor suppressor: normal gene involved in suppressing cell cycle
- What would be the result of a mutation in tumor suppressor gene?

Gene	Familial Cancer Syndrome	Protein Function	Sporadic Tumors with Mutations
RB1	Hereditary retinoblastoma	Transcriptional co-repression	Sporadic retinoblastoma, osteosarcoma, small-cell lung carcinoma (SCLC), breast carcinoma, bladder carcinoma
TP53 (p53)	Li-Fraumeni syndrome	Transcription factor	>50% of all cancers
WT1	Wilms tumor (nephroblastoma)	Transcription factor	Pediatric kidney cancer
BRCA1	Familial breast and ovarian cancer	DNA repair, cell cycle control (genomic stability)	Unknown
BRCA2	Familial breast and ovarian cancer	DNA repair (homologous recombination, genomic stability)	Unknown

Two-hit hypothesis

- Also known as Knudson's [two-hit hypothesis](#)
 - Proposed by Alfred Knudson
 - Both alleles of a tumor suppressor gene must be inactivated for cancer to develop.
 - This typically involves two “hits” or mutations
- First Hit – Often a germline mutation inherited from a parent
- Second Hit – A specific hit in a somatic cell
- Genes involved are typically [recessive](#) in nature
- Loss-of-heterozygosity is involved



Retinoblastoma

- Retinoblastoma is a rare, malignant tumor of the retina, primarily affecting young children.
- 1:20,000 births
 - Example of tumor suppressor and first to support two-hit hypothesis
- *RB1* gene
- Retinoblasts normally stop dividing after embryogenesis
- Results in retinal tumors (intraocular)
- 90% penetrance



	Bilateral	Unilateral	Total
Hereditary	25%–30%	10%–15%	35%–45%
Nonhereditary	0	55%–65%	55%–65%
Total	25%–30%	70%–75%	100%

Fig. 15-7, Thompson & Thompson Genetics in Medicine, 8th Ed.
Chial, H. (2008) Tumor suppressor (TS) genes and the two-hit hypothesis. Nature Education 1(1):177

RB Two-hit hypothesis

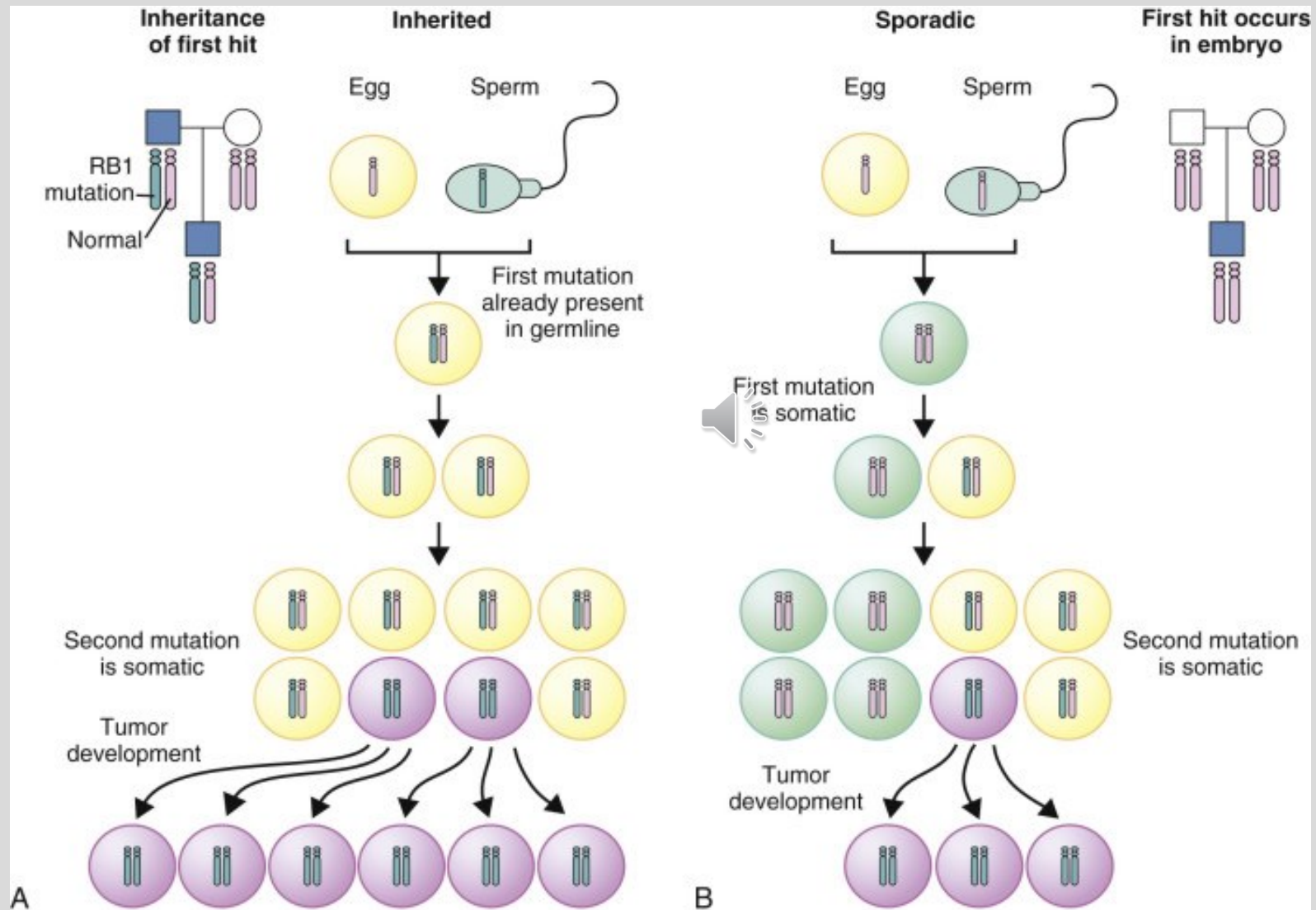


Figure 11-4, Medical Genetics, 5th Ed

Familial breast and ovarian cancer

- *BRCA1*, *BRCA2* genes
- Only 1-3% breast cancer due to *BRCA1* or *BRCA2* mutations

Cancer Type	General Population Risk	Cancer Risk When Mutation Present	
		<i>BRCA1</i>	<i>BRCA2</i>
Breast in females	12%	50%-80%	40%-70%
Second primary breast in females	3.5% within 5 yr	27% within 5 yr	12% within 5 yr
	Up to 11%		40%-50% at 20 yr
Ovarian	1%-2%	24%-40%	11%-18%
Male breast	0.10%	1%-2%	5%-10%
Prostate	15% (N. European origin)	<30%	<39%
	18% (African Americans)		
Pancreatic (both sexes)	0.50%	1%-3%	2%-7%

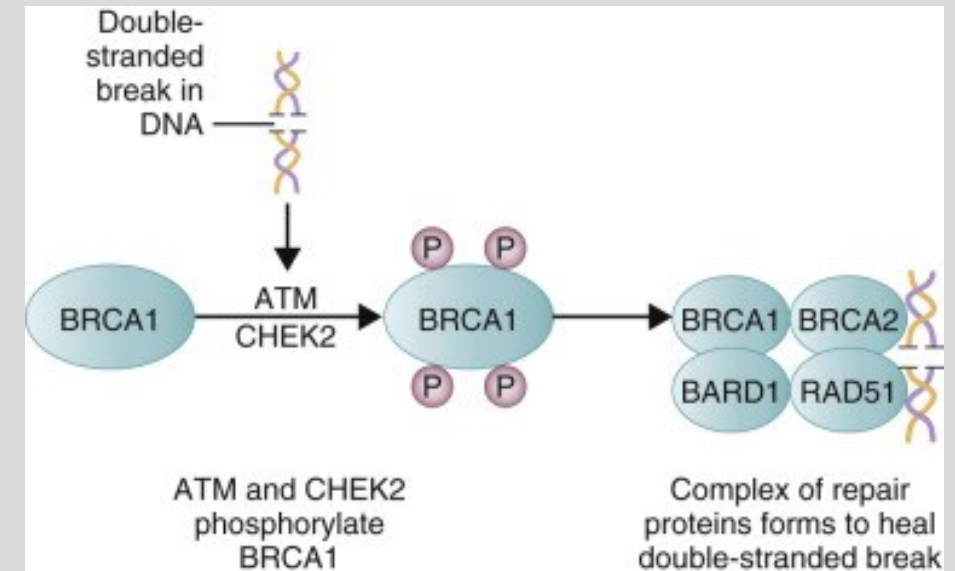
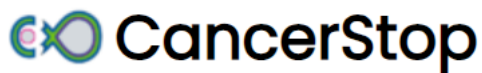


Fig. 11-10, Medical Genetics, 5th Ed

Other genes with inherited/germline pathogenic mutations can also lead to breast cancer

- **ATM**: DNA Repair gene (or kills cell that can't be fixed). 2 mutated copies causes **Ataxia-telangiectasia**; 1 mutated copy linked high rates of breast cancer in some families.
- **PALB2**: Interacts with *BRCA2* protein. Mutation can lead to a higher risk of breast cancer.
- **TP53**: Stops growth of cells with damaged DNA. Inherited/germline mutations cause **Li-Fraumeni syndrome**.
- **CHEK2**: DNA repair gene. *CHEK2* mutation increases breast cancer risk.
- **PTEN**: Regulates cell growth. Inherited mutations can cause **Cowden syndrome**.
- **CDH1**: Inherited mutations cause **hereditary diffuse gastric cancer**. Females with *CDH1* mutations have an increased risk of invasive lobular breast cancer.
- **STK11**: Inherited *STK11* mutations can lead to **Peutz-Jeghers syndrome**.

Inherited mutations in several other genes have also been linked to breast cancer, but these account for only a small number of cases.



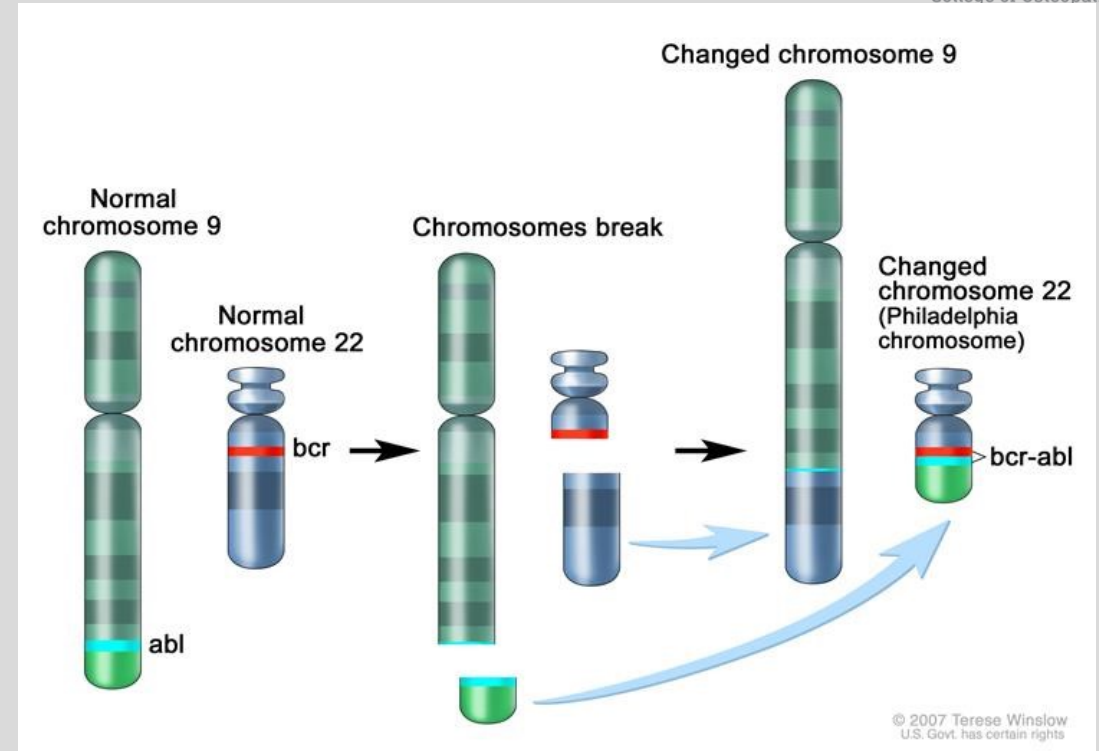
<https://www.cancerstop.dev/genes-and-more>

American Cancer Society:

<https://www.cancer.org/cancer/types/breast-cancer.html>

Sporadic cancers-oncogenes

- Inherited cancers only account for small %
- Ras proto-oncogene
 - Small GTPase involved in signal transduction
- Ras mutations found in 30% all cancers
- Philadelphia chromosome
 - Translocation results in *BCR-ABL1* fusion
 - ABL1 tyrosine kinase-becomes overactive with fusion
 - Chronic myelocytic leukemia (CML) and 25% of acute lymphoblastic leukemia (ALL)



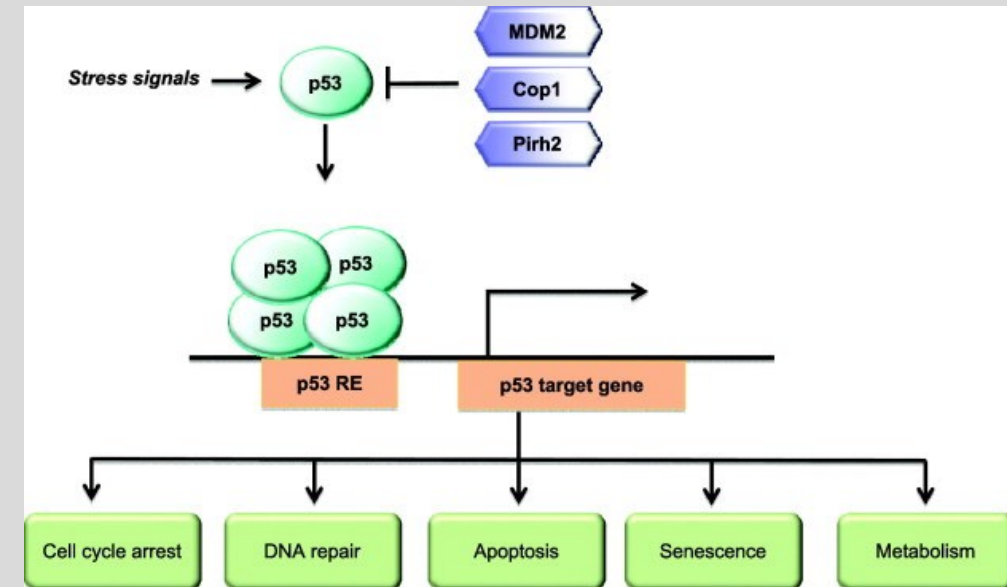
National Cancer Institute, NIH



<https://www.cancerstop.dev/genes-and-more>

Sporadic cancers-tumor suppressor p53

- Rare dominant inherited mutation of p53 - Li-Fraumeni syndrome
- >50% sporadic cancer include mutations in p53
- p53 expression induced by DNA damage
 - Can induce apoptosis to stop proliferation of damaged cells
- Specific nucleotide alterations in p53 shown with benzopyrene
 - Mutagen in cigarette smoke, direct link to lung cancer



Liu et al, (2015) Tumor suppressor p53 and its mutants in cancer metabolism. *Cancer Letters* 356(2): 197

Environmental risk factors

- Contribute to multifactorial inheritance
- Smoking:
 - Approx. 30% all cancer deaths
 - 55 known/suspected carcinogens
- Vaping: Increases risk for asthma, chronic bronchitis, emphysema, and COPD
(<https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2772829>)
- Diet:
 - 14-30% cancer deaths attributed to overweight and obesity
 - Associated risk increased for cancer such as colon, breast, kidney, liver
- Occupation



Occupation	Carcinogen Exposure	Associated Cancer Type
Iron and steel founding	PAH, chromium, nickel, formaldehyde	Lung
Copper mining and smelting	Arsenic	Skin, bronchus, liver
Underground mining	Radon (ionizing radiation)	Lung
Aluminum production	PAH	Lung
Coke production	PAH	Lung, kidney
Painting	Chromium, solvents	Lung
Furniture and cabinet making	Wood dust	Nasal sinus
Boot and shoe manufacture	Leather dust, benzene	Nasal sinus, leukemia
Rubber industry	Aromatic amines, solvents	Bladder, leukemia
Nickel refining	Nickel	Nasal sinus, bronchus
Vinyl chloride manufacture	Vinyl chloride	Liver
Dye and textile production	Benzidine-based dyes	Bladder

Table 7-2, The Molecular Basis of Cancer, 4th Ed

EPA data on diseases and environmental contributions



<https://www.epa.gov/report-environment/disease-and-conditions>

Summary

- What are the characteristics of multifactorial inheritance
- Qualitative and quantitative traits and how they relate to multifactorial disorders
- Understand empirical risk and threshold model
 - Clinical examples as models to estimate empirical risk (cleft palate, pyloric stenosis; etc)
- Know examples of multifactorial disorders (NTD, hypertension, diabetes)
- Relate genetic predisposition to and genetic mechanisms leading to cancer
 - Proto-oncogene to oncogene. What are the functions of oncogenes and tumor suppressors? What types of mutations are needed to create each?
 - Retinoblastoma, inheritance, and risk. 2-hit hypothesis.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>